thimble we stop the rotational movement of thimble, when we hear click sound.

Least Count of Metric Micrometer

When we rotate the thimble to one rotation, the distance moved by the spindle is called 'pitch' of the micrometer. Thimble scale is having 50 divisions itself. when we rotate the thimble to one rotation, the spindle moves 0.5mm

Formula

$$\text{Least Count of metric Micrometer} = \frac{\text{pitch}}{\text{Total no of thimble Divisions}}$$

$$= \frac{0.5}{50} = \frac{5}{500} = \frac{1}{100}$$

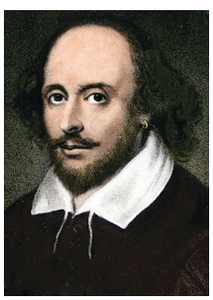
Least Count of metric Micrometer = 0.01 *mm*

Least Count of British Micrometer

In British micrometer, main scale (Sleeve) is having 40 threads in one inch. When we rotate thimble to one rotation, spindle moves to 1/40 inch. Thimble scale is having 25 divisions itself.



William Gascoigne



William Gascoigne was an English astronomer, mathematician and maker of scientific instruments from Middleton, America Leeds who invented the micrometer.

$$\text{Least Count of metric Micrometer} = \frac{\text{pitch}}{\text{Total no of thimble Divisions}}$$

$$= \left(\frac{1/40}{25}\right)$$

$$= \frac{1}{40} \text{X} \frac{1}{25}$$

$$= \frac{1}{1000}$$

Least count of British Micrometer = 0.001 *inch*

METHOD OF MEASURING IN METRIC MICROMETER

Errors in Micrometer

There are two types of Error

1. Positive Error
2. Negative Error

POSITIVE ERROR

When the micrometer is closed if the zero of the thimble is not touch the zero of sleeve, the error is said to be positive. The number of thimble divisions between the zero of the thimble divisions between the zero of thimble and zero of sleeve is the positive error should be subtracted from the original reading of micrometer.

Problem

When we measure an object, the micrometer shows 15.12 mm. The positive error of the micrometer so 0.03 mm .What is the correct dimension of the object?

Micrometer Reading	=	15.12 mm
Positive Error Reading	=	(–) 0.03 mm
Correct Dimension of Object		15.09 mm

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